

Error-type	Error-type-finegrained	Definition
Incorrect references	Invalid Reference, Concurrent work	The reviewer is not citing the appropriate sources (non peer-reviewed or concurrent work) in the current statement.
Less rigor in reviewing methodology and experiments	Out-of-scope, Invalid Criticism, Less rigor in reviewing methodology and experiments	The reviewer is suggesting things beyond the scope of the paper or the reviewer's criticism is invalid.
Misinterpretation of claims and ideas presented in the paper	Misunderstanding, Neglect, Inexpert Statement, Unstated statement	The reviewers is misinterpreting the claims and ideas presented in the paper and overlooked important details of the paper or the reviewer is exhibiting lack of domain knowledge or not supported by the content of the paper.
Superficial and vague review	Misinterpret Novelty, Vague Critique, Subjective, Superficial Review, Missing Reference	In the current statement, the reviewer has misinterpreted novelty or the reviewer is lacking specificity of the components.
Incomplete, incorrect, or copied summary	Inaccurate Summary, Summary Too Short, Copy-pasted Summary	The summary is misrepresenting the content of the paper, or too short or directly copied from the paper.
Syntactic, structural, or semantic issues in the review	Typo, Contradiction, Misplaced attributes, Duplication	The review segment has typological errors that are affecting the clarity.
Misidentification of Syntactic or structural issues in the paper	Writing, Misunderstanding of the Submission Rule	The reviewer has misidentified the structural issues in the paper.

Table 3: Error type and their definitions

Error-Type	Possible set of Rebuttal Actions
Incorrect references	Reject Request
Less rigor in reviewing methodology and experiments	Accept for future work, Reject criticism, Refute question
Misinterpretation of claims and ideas in the paper	Contradict Assertion, Refute question, Reject criticism
Superficial and vague review	Refute question, Reject criticism, Contradict assertion
Incomplete, incorrect or copied summary	All
Syntactic, structural, and semantic issue in the paper	Reject criticism
Non-deficient	Answer question, The task is done, The task will be done, Concede criticism, Mitigate criticism, Accept praise

Table 4: Static mapping of error-type to rebuttal actions

Deficiency status	Question
True	The review statement is not valid. It contains either factual errors, lacks constructive feedback, is subjective, or is without evidence (Deficient).
False	The review statement is valid in terms of factuality and constructive feedback (Non-deficient).

Table 5: Natural language statement mapping for deficiency status

Error-type	Statement
Incorrect references	The reviewer is not citing the appropriate sources (non peer-reviewed or concurrent work) in the current statement.
Less rigor in reviewing methodology and experiments	In the current statement, the reviewer is suggesting things beyond the scope of the paper or the reviewer’s criticism is invalid.
Misinterpretation of claims and ideas presented in the paper	In the current statement, the reviewer is misinterpreting the claims and ideas presented in the paper and overlooked important details of the paper or the reviewer is exhibiting lack of domain knowledge or not supported by the content of the paper.
Superficial and vague review	In the current statement, the reviewer has misinterpreted novelty or the reviewer is lacking specificity of the components. Do you agree?
Incomplete, incorrect, or copied summary	In the current statement, the summary is misrepresenting the content of the paper, or too short or directly copied from the paper.
Syntactic, structural, or semantic issues in the review	The current review statement has typographical errors that are affecting the clarity.
Misidentification of syntactic or structural issues in the paper	In the current review statement, the reviewer has misidentified the structural issues in the paper.

Table 6: Natural language statement mapping for error types

Rebuttal action	Statement
Reject work	The request needs to be rejected.
Accept for future work	The suggested things needs to be accepted as future work.
Reject criticism	The criticism needs to be rejected.
Refute question	The question needs to be disproved.
Contradict assertion	The statement needs to be contradicted.
Mitigate criticism	The rebuttal statement needs to be generated in a manner that represents the statement is not important.
Answer question	The question needs to be answered.
The task is done	The rebuttal statement needs to specify the task has already been done and pinpoint where.
The task will be done	The rebuttal statement needs to specify the task will be done in camera ready.
Concede criticism	The rebuttal statement needs to admit to the provided criticism.
Accept praise	The rebuttal statement needs to accept the praise.

Table 7: Natural language statement mapping for rebuttal actions

The screenshot shows a web interface for a tool called "DEFEND: Automated Rebuttals for Peer Review". The interface is contained within a light blue rounded rectangle. At the top, the title "DEFEND: Automated Rebuttals for Peer Review" is displayed in a bold, dark blue font. Below the title, there are three main input sections. The first section, labeled "Paper Title:", contains a single-line text input field. The second section, labeled "Upload Paper (PDF):", contains a file upload button labeled "Choose file" and the text "No file chosen". The third section, labeled "Review:", contains a large, multi-line text area for writing a review. At the bottom left of the form, there is a blue button with the text "Generate Rebuttal".

Figure 2: Defend: Screenshot of the tool

Deficiency Prediction

Given a paper submission and its corresponding review, your job is to assess the deficiency of a given segment. You need to assess if each segment/sentence of the review is "deficient" or not.

The criteria for deficiency are: {DEFICIENT}

Deficient segments can have following error types (All possible error types with definitions are given below): {ERROR_TYPES}

Paper title: {PAPER_TITLE} Paper content: {PAPER_CONTENT} Review: {REVIEW}

Segment to be predicted: {SEGMENT_TO_BE_PREDICTED}

You are supposed to predict whether the given segment is deficient or not. Please take care of the following while predicting for the given segment 1. Your answer should only contain [Yes/No] and nothing else. 2. Please consider paper title, paper content, content of the entire review, criteria for deficiency, definition of error-types and general domain knowledge.

Error-type Prediction

Given a paper submission, its corresponding review your job is to identify the error-type of a segment. Some important information is provided below

The criteria for deficiency are: {DEFICIENT}

Deficient statements can have following error types (All error types with definitions are given below): {ERROR_TYPES_DEFINITIONS}

Paper title: {PAPER_TITLE} Paper content: {PAPER_CONTENT} Review: {REVIEW}

Current segment: {SEGMENT} Deficiency status: {DEFICIENCY_STATUS} Strictly select from the list of error-types and show error-type only and nothing else. If there is no valid error type or the review segment is not deficient return None and nothing else

Rebuttal action prediction

Given a paper submission, its corresponding review your job is to predict a rebuttal action for a particular segment.

The criteria for deficiency are: {DEFICIENT}

Deficient statements can have following error types (All error types with definitions are given below): {ERROR_TYPES}

Definition of rebuttal actions: {REBUTTAL_ACTIONS_DEFINITIONS}

Paper title: {PAPER_TITLE} Paper content: {PAPER_CONTENT} Review: {REVIEW}

Current segment to be predicted: {SEGMENT_TO_BE_PREDICTED}

Deficiency status of the current segment: {DEFICIENCY_STATUS} Error type of the current segment: {ERROR_TYPE} Subset of the rebuttal actions applicable for the particular segment: {SUBSET_REBUTTAL_ACTIONS}

Please select an appropriate rebuttal action for the current segment. Please take care of the following while predicting for the given segment 1. Only output one rebuttal action from the subset of the rebuttal actions provided and nothing else. 2. Consider paper title, paper content, content of the entire review, criteria for deficiency, definition of error-types, deficiency status, and error-types for the current segment, and general domain knowledge.

Table 8: Prompts

Consolidation

Given a paper title, paper content, review, and rebuttal segments for each review segment.

Paper title: PAPER_TITLE Paper content: PAPER_CONTENT Review and rebuttal information for each segment: {REVIEW_REBUTTAL}

Please consolidate the entire rebuttal in a structured format replying to each review segment to send back to the reviewer to increase chances of acceptance.

While consolidating please take care of the following: 1. Consider the provided review segment with their rebuttal and use them to construct a coherent rebuttal. 2. Do not provide strategy to construct a rebuttal in the rebuttal but provide the rebuttal only. Do not write Review Segment 1 etc in the final rebuttal. Just the review segment in bold is sufficient. 3. Do not include segments with No Rebuttal Needed in consolidating the final rebuttal. 4. Provide a single statement for thanking the reviewers for their valuable feedback. Do not provide a thanking statement with each and every rebuttal segment. 5. The consolidated rebuttal should look like an original author's response without any redundant thanking statements or repetitions. 6. Provide a single statement for thanking the reviewers for their valuable feedback in the start of the rebuttal. 7. The generated rebuttal should be in markdown format. 8. The rebuttal generated should not use verbose or confusing language. 9. The rebuttal generated should be relevant to the points in the review.

Direct rebuttal generation

You are a rebuttal writing assistant for academic peer reviews.

Given the full review, paper title, and paper content, generate a complete rebuttal addressing all the reviewer comments.

Paper title: {PAPER_TITLE} Paper content: {PAPER_CONTENT}

Review: {FULL_REVIEW}

Please take care of the following while generating rebuttal: 1. Generate a rebuttal in fluent academic professional tone addressing the concerns and suggestions in the review. It should not contain any offensive terms. 2. Avoid generic statements. Be precise and point-wise. 3. The rebuttal generated should not use verbose or confusing language. 4. The rebuttal should be complete. Please take care of covering all the required points from the review that needs to be answered. 5. All the responses should be factually and logically correct, detailed, and supported by evidence. 6. The rebuttal generated should be relevant to the points in the review. 7. The rebuttal generated should be constructive, either it should foster discussion or offer solutions. It should provide actionable feedback. 8. The rebuttal generated should help the author in getting their scores increased and should help in increasing the chances of acceptance. 9. The rebuttal generated should be consistent and coherent and should not have any contradicting statements. 10. Consider paper title, paper content, and general domain knowledge while generating a rebuttal for the review. 11. Be very concise in generating the rebuttal. 12. For a review segment which is summary of the paper or just a statement about the paper, there is no rebuttal needed for those review segments. 13. Do not include phrases like "here's a detailed rebuttal addressing the reviewer's comments", Rebuttal to Reviewer Comments etc. in the rebuttal. Directly start from a thanking statement.

Table 9: Prompts

Segment-wise rebuttal generation

You are assisting in academic peer-review rebuttal writing.

Given the current review segment, paper title, and paper content generate a rebuttal segment specifically for the current review segment.

Paper Title: {PAPER_TITLE} Paper Content: {PAPER_CONTENT} Current review segment to be predicted: {SEGMENT}

Please generate a rebuttal segment for the current review segment. If no rebuttal is needed output "No rebuttal needed".

Please take care of the following while generating the rebuttal segment: 1. A part of the rebuttal is to be generated only for the current review segment to be predicted. 2. Generate a rebuttal in fluent academic professional tone addressing the concerns and suggestions in the review. It should not contain any offensive terms. 3. Avoid generic statements. Be precise and point-wise. 4. The rebuttal segment generated should not use verbose or confusing language. 5. The rebuttal segment should be complete. Please take care of covering all the required points from the review segment that needs to be answered. 6. All the responses should be factually and logically correct, detailed, and supported by evidence. 7. The rebuttal segment generated should be relevant to the points in the review segment. 8. The rebuttal segment generated should be constructive, either it should foster discussion or offer solutions. It should provide actionable feedback. 9. The rebuttal segment generated should help the author in getting their scores increased and should help in increasing the chances of acceptance. 10. The rebuttal segment generated should be consistent and coherent and should not have any contradicting statements. 11. Consider paper title, paper content, and general domain knowledge while generating a rebuttal segment for the current review segment. 12. Be very-very concise in generating a rebuttal segment. Try to provide a concise, short and complete answer. 13. If the review segment is a summary or just a statement about the paper, the rebuttal segment will be "No rebuttal needed". 14. For the review segments where there is no need to respond, please provide the output as no rebuttal needed.

Table 10: Prompts

Rebuttal Generation-SA You are assisting in academic peer-review rebuttal writing. Given the current review segment, paper title, and paper content generate a rebuttal segment specifically for the current review segment.

The criteria for deficiency are: {DEFICIENT}

Deficient statements can have following error types (All error types with definitions are given below): {ERROR_TYPES}

Definition of rebuttal actions: {REBUTTAL_ACTIONS_DEFINITIONS}

Paper title: {PAPER_TITLE} Paper content: {PAPER_CONTENT}

Current review segment to be predicted: {SEGMENT_TO_BE_PREDICTED}

Deficiency status of the current segment: {DEFICIENCY} Error type of the current segment: {ERROR_TYPE} Rebuttal action for the current segment: {REBUTTAL_ACTION}

Please generate a rebuttal segment for the current review segment. If no rebuttal is needed output "No rebuttal needed".

Please take care of the following while generating the rebuttal sentence: 1. A part of the rebuttal is to be generated only for the current review segment to be predicted. 2. Generate a rebuttal in fluent academic professional tone addressing the concerns and suggestions in the review. It should not contain any offensive terms. 3. Avoid generic statements. Be precise and point-wise if needed. 4. The rebuttal segment generated should not use verbose or confusing language. 5. The rebuttal segment should be complete. Please take care of covering all the required points from the review segment that needs to be answered. 6. All the responses should be factually and logically correct, detailed, and supported by evidence. 7. The rebuttal segment generated should be relevant to the points in the review segment. 8. The rebuttal segment generated should be constructive, either it should foster discussion or offer solutions. It should provide actionable feedback. 9. The rebuttal segment generated should help the author in getting their scores increased and should help in increasing the chances of acceptance. 10. The rebuttal segment generated should be consistent and coherent and should not have any contradicting statements. 11. Consider paper title, paper content, criteria for deficiency, definition of error-types, deficiency status, error-types, rebuttal action for the current segment, and general domain knowledge. 12. Be very concise in generating a rebuttal segment. 13. If the review segment is a summary or just a statement about the paper, the rebuttal segment will be "No rebuttal needed". 14. For the review segments where there is no need to respond, please provide the output as no rebuttal needed.

Table 11: Prompts

You are an academic assistant. You are given:

1. A REVIEW SEGMENT written by a peer reviewer.
2. A TARGET PAPER containing its title, abstract, and full text (section-wise).

Your task is to identify all paragraphs or passages in the paper that are most relevant (in answering the questions, clarifying doubts, or just supporting the claims) to the review segment.

Guidelines: - Return only text copied directly from the given paper. Do not rewrite or paraphrase. - Select the smallest set of passages that meaningfully support, contradict, or relate to the review segment. - Prefer passages with high semantic overlap, methodological relevance, or matching claims. - If nothing relevant exists, state: "No directly relevant evidence found." - Section header can be a title summarizing the paragraph not necessarily from the paper.

Output Format: <Relevant content with section titles>

REVIEW SEGMENT: {REVIEW_SEGMENT} PAPER CONTENT: {PAPER_CONTENT}

Table 12: Prompts

You are assisting in academic peer-review rebuttal writing.

Does the provided rebuttal segment contains any refutation of the review segment? You are provided with review segment, predicted rebuttal segment, and the rebuttal action. Review segment: {REVIEW_SEGMENT} Rebuttal segment: {PREDICTED_REBUTTAL_SEGMENT}

Refutation: A refutation is a direct response to a criticism, claim, or argument that seeks to disprove, counter, or weaken it. Provide an explanation of why it is a refutation and then provide an answer as "Yes", or "No". Answer is "Yes" if it has full refutation, "No" for no refutation.

Please take care of the following together with the above definition while predicting whether their is a refutation or not

The rebuttal segment is not a refutation - If it is acknowledging the issue and resolving or acknowledging and providing clarification of the doubt. - If it is providing a clarification to the question, or a doubt in the review segment, even if it contains a negation. - If it is resolving the reviewer's confusion. - If it is doing what the reviewer has suggested. - If it is directly addressing the reviewer's comments in a positive way. - If it is mitigating the criticism. - If it agrees with the reviewer's claim, acknowledges the issue, promises to revise or clarify, or reframes the result without disputing the reviewer's point. - If it is adding extra work to mitigate the criticism then it is not a refutation. - If it is adding evidence that strenthens the paper without completely rejecting the criticism. - If it is specifying something that the required task will be done or will be included in camera-ready. - If the review segment is a question or doubt. - If the review segment is summary of the paper, and rebuttal segment is accepting or not replying to it saying "No rebuttal needed".

The rebuttal segment is a refutation: - If the review segment is a misunderstanding/misconception/wrong information, then it is a refutation except for the situation when the review segment is a question or doubt. - If it is countering an argument in the review segment that undermines, weakens, or disproves the claim in the review segment.

Strictly follow the provided output format. Your output must follow the following output format. Please take care of providing the output in one line having explanation and refutation separated by a pipe character.

Output format: Explanation: explanation | Refutation: refutation

Table 13: Prompts

You are assisting in academic peer-review rebuttal writing.

You are assisting in academic peer-review rebuttal writing.

You will be provided with three pieces of text: Review Segment: The original point or question from a peer reviewer. Gold Rebuttal Segment: A high-quality, correct rebuttal to the review segment. Generated Rebuttal Segment: The rebuttal that you need to fact-check.

Review Segment: {REVIEW_SEGMENT} Gold Rebuttal Segment: {GOLD_REBUTTAL_SEGMENT} Generated Rebuttal Segment: {PREDICTED_REBUTTAL_SEGMENT}

Is the generated rebuttal segment consistent across various points and does not have any contradiction as compared to the gold one?

Evaluation guidelines: - If the review segment provides summary of the paper "accept praise" or "no rebuttal needed" both are acceptable and is considered consistent. - The generated rebuttal segment is consistent, if it addresses the same issue raised by the review (not off-topic). - The generated rebuttal segment is consistent, if its internal reasoning is coherent and free of contradictions.

Your output should only contain yes/no together with an explanation and nothing else. Explanation and Consistency both should be separated by a pipe character. Your output must follow the following output format.

Output format: Explanation: explanation | Consistency: consistency

Table 14: Prompts

You are assisting in academic peer-review rebuttal writing.

You will be provided with four pieces of text:

Review Segment: The original point or question from a peer reviewer. Gold Rebuttal Segment: A high-quality, correct rebuttal to the review segment. Gold Rebuttal Action: It reflects what kind of action has been taken in gold rebuttal segment. Generated Rebuttal Segment: The rebuttal that you need to fact-check.

Review Segment: {REVIEW_SEGMENT} Gold Rebuttal Segment: {GOLD_REBUTTAL_SEGMENT} Gold Rebuttal Action: {GOLD_REBUTTAL_ACTION} Generated Rebuttal Segment: {PREDICTED_REBUTTAL_SEGMENT}

Your task is to determine if the generated rebuttal segment is factually correct given the review and gold rebuttal segment.

Following the following process to examine the factual correctness of the generated rebuttal segment.

1. Carefully read the Review Segment to understand the reviewer's concern or question. 2. Use the Gold Rebuttal Segment and gold rebuttal action to identify the key factual claims and evidence it uses to address the reviewer's point. Also note any specific data, table references, or logical arguments presented.

Evaluation guidelines: - Strong evidence/justifications are not always needed. Only penalize when it is needed and present in the gold rebuttal segment. - Do not penalize if the rebuttal segment omits detailed evidence where the gold rebuttal segment also does not have it. - Penalize when claims are unsupported, misleading, or incorrect relative to the gold rebuttal segment. - If the review segment provides summary of the paper "accept praise" or "no rebuttal needed" both are acceptable and gets high score for factual correctness. - Be strict about factual alignment, and focus only on correctness and completeness (not style). - All the factual claims present in gold rebuttal segment should be covered in generated rebuttal segment for it to be factually correct. - Pay attention to whether it accurately reflects the same facts as the gold rebuttal segment (e.g., added results, clarifications, actions taken). If it accurately reflects the same facts as the gold rebuttal segment then only it is factually correct.

Your output should only contain yes/no together with an explanation and nothing else. Explanation and Factual correctness both should be separated by a pipe character. Your output must follow the following output format.

Output format: Explanation: explanation | FC:factual_{correctness}

Table 15: Prompts

Create partition/segments of the review based on different sections or meaningful units.

Review: {Review} While generating please take care of the following: - Segments are a meaningful continuous part from the review. - Segments should be from the review itself, no content should be added extra in the segments. - There should be no overlap of sentences between two segments. - The segments should be separated by | character. - Summary should be in one single segment. - Each component of strengths and weaknesses should be separate segments. - Each question should be a separate segment. - All segments combined should constitute the complete review.

Table 16: Prompts

Review: Paper presents an instance weighting method that alters the loss function of unsupervised MT pipeline at back-translation phase. The proposed method is incorporated in the fine-tuning phase of the unsupervised MT pipeline and makes use of round-trip translation BLEU score of to change the loss weight for a sequence where the scores are normalized over the batches. Method is applied on two pre-trained unsupervised MT systems (XLM and MASS based) and shows some statistically significant gains over XLM based pre-training.

Rebuttal: No rebuttal needed

Accept

Refine

Review: Simplicity and practicality are the strengths of the paper.

Rebuttal: No rebuttal needed

Accept

Refine

Review: Within the framework provided by the paper, several other filtering/selection/weighting methods can be benchmarked.

Rebuttal: We appreciate the reviewer's suggestion to benchmark our approach against other filtering/selection/weighting methods. We agree that this would provide a more comprehensive evaluation. In the revised version, we will include a discussion of how our method compares to other existing techniques for back-translation filtering, specifically those mentioned in related work such as uncertainty-based confidence estimation (Wang et al., 2019a) and co-curricular learning (Wang et al., 2019b). Furthermore, we will add experiments using a combination of round-trip BLEU with other quality estimation metrics such as language model scores, inspired by Dou et al. (2020), to explore whether a hybrid approach can yield further improvements. We believe these additional experiments will strengthen the paper and provide a more thorough analysis of our proposed method.

Accept

Refine

Figure 3: Defend: Screenshot of the tool

Review: More details could be provided. The epoch size is set to 2M, is this number of tokens processed per epoch, update steps to be taken or else?

Rebuttal: We thank the reviewer for pointing out the missing details. The epoch size of 2,000,000 refers to the number of update steps taken during training. We will clarify this in the revised manuscript.

Assistant: The review statement is valid in terms of factuality and constructive feedback (Non-deficient). Do you agree? Reply in Yes/No.

Type your reply...

Send

Accept

Refine

Figure 4: Defend: Screenshot of the tool

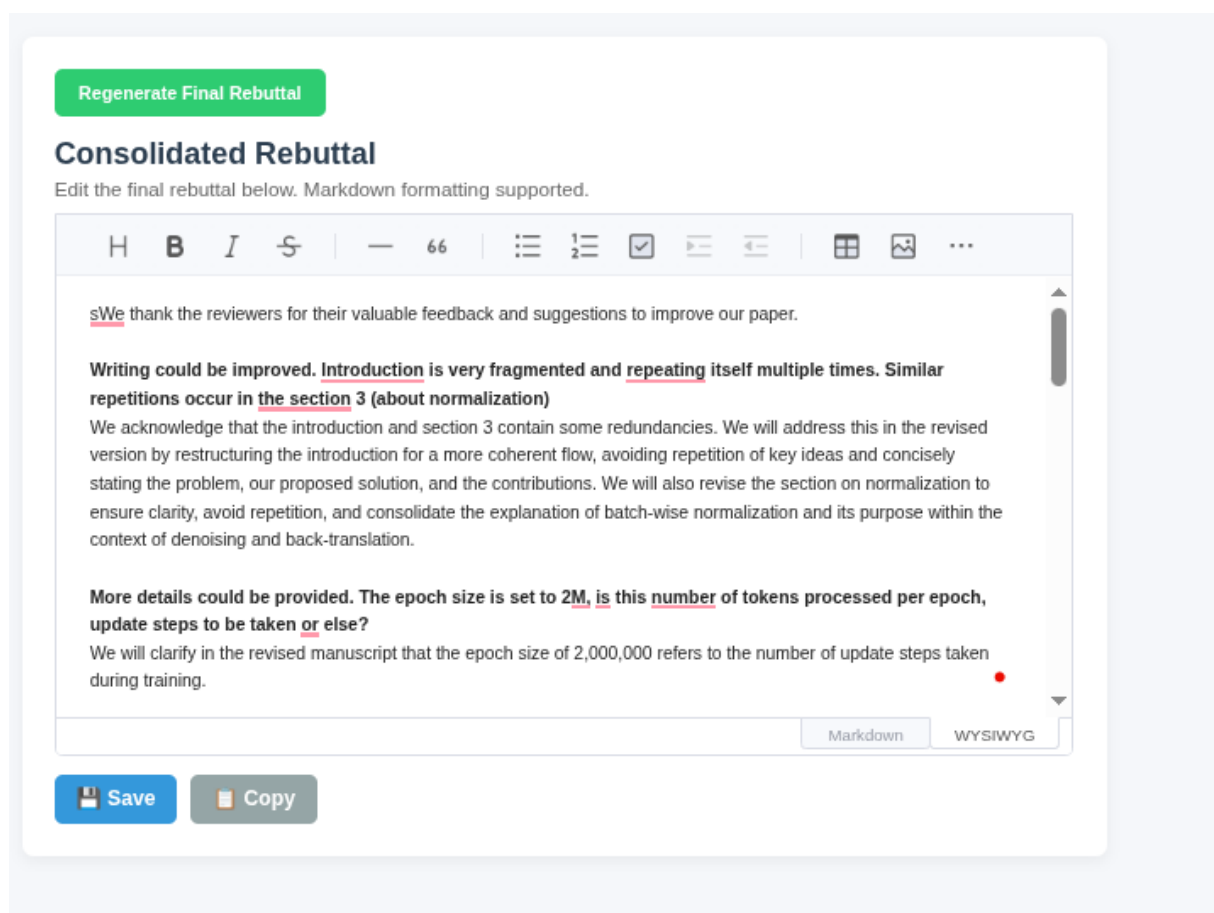


Figure 5: Defend: Screenshot of the tool